

Virender Kumar

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Professional Summary

Results-driven **Software Engineer II** with 2+ years of experience in **C/C++ backend development**, designing and optimizing **scalable, high-performance systems** on **Linux**. Skilled in **network programming**, **secure protocols (TLS, SSL)**, and **web technologies (HTTP, DNS, TCP/IP)**. Experienced in **CI/CD pipeline development**, **automation**, **log analysis**, and working with **cloud platforms (AWS, GCP, Azure)**. Adept at solving complex concurrency, scalability, and reliability challenges across distributed environments and driving DevOps process improvements.

Professional Experience

Software Engineer II

Teradata

Jun 2025 – Present

- Designed and implemented high-performance components in **C++** for distributed backup systems, running on **Linux**.
- Implemented **TLS 1.3** protocol in C++ data streaming client, reducing handshake latency by **40%** and improving network security.
- Improved **throughput and scalability** by 30% through multithreading and socket-level optimizations in the Network Client.
- Enhanced reliability of data transfer over **HTTP/TCP/IP**, enabling faster and more resilient cloud-based operations.
- Developed data immutability and locking mechanisms to secure **10TB+ critical user data** across AWS S3, Azure Blob, and GCS.
- Implemented **token bucket rate limiter** and connection retry logic to prevent overloads in large-scale network communication.
- Investigated and resolved a **thread deadlock issue** during a customer call, provided a live workaround, and delivered a production hotfix within 2 days.
- Collaborated with Dell engineering to integrate **PPDM agent**, ensuring smooth enterprise adoption of the distributed backup platform.
- Integrated the **MCP server** with core product for cross-system automation; project advanced to internal architectural review.
- Acted as the **Point of Contact (POC)** for **DevOps and CI/CD**, maintaining and enhancing build and deployment pipelines across teams.
- Designed and deployed a new CI/CD pipeline for the **SLESSP7 architecture** with updated **CMEK (Customer-Managed Encryption Key)** requirements, improving build reliability and security compliance.
- Automated monitoring, log parsing, and release workflows using **Python and Bash**, improving operational efficiency by 30%.

Associate Software Engineer

Teradata

Jul 2023 – Dec 2024

- Developed and optimized core **C++ services** for backup and data management in distributed environments.
- Enhanced application performance for S3-compatible object stores (AWS, Dell EMC, IBM COS, MinIO).
- Designed and tested **rate-limiting** and retry mechanisms to prevent congestion and DDoS-like load scenarios.

- Worked on data **log processing and analysis tools** in Python to track performance metrics and detect failures.
- Mentored and guided a team of **3 junior engineers**, providing technical support in debugging and feature development.
- Collaborated across backend, DevOps, and QA teams to manage 50TB+ distributed storage environments.

DevOps Engineer Intern

Teradata

Jan 2023 – Jun 2023

- Built and automated **CI/CD pipelines** and infrastructure provisioning across AWS, GCP, and Azure.
- Developed **Python and Bash scripts** for deployment, log monitoring, and system diagnostics.
- Improved pipeline efficiency by 30% through automation of Linux-based environments.

Technical Skills

- **Programming Languages:** C, C++, Python, Java, Bash, JavaScript
- **Core Expertise:** Backend Development, Multithreading, Network Programming, Distributed Systems, Log Processing
- **Protocols & Security:** TCP/IP, SSL/TLS, DNS, HTTP, SAML, OIDC, Web Security (DDoS Mitigation, App Security)
- **DevOps & CI/CD:** Jenkins, Docker, CMake, Git, Linux/Unix, Build Automation, CMEK Integration
- **Cloud Platforms:** AWS, GCP, Azure (S3, Blob Storage, GCS)
- **Databases:** MySQL, PostgreSQL
- **Software Development:** OOP, Data Structures & Algorithms, Performance Optimization, SDLC, Agile

Projects

MCP Server Integration (Hack Day Winner)

- Integrated MCP Server with Teradata's enterprise product for distributed job orchestration and monitoring.
- Designed scalable automation tools; recognized during innovation hack day and advanced to architecture review.

AI Agent Integration

- Prototyped Python-based AI agent integrated with product API for automated data insights.
- Developed tools for data preprocessing, log analysis, and intelligent alerting.

Education

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Graphic Era Hill University, Dehradun

CGPA: 8.64 / 10.0

Graduated: 2023